Statistical Analysis of Public Awareness of Groundwater Contamination from the Northrop Grumman Plume Affecting Select Communities in Nassau County, New York

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Abstract

The Grumman Aircraft Engineering Corporation of Bethpage, NY was a booming aeronautical business in the early 1930s. After many years of reviewing historical records involving production practices at Northrop Grumman, there is evidence that a groundwater plume developed due to disposal of waste liquids and leakage from solvent bulk storage containers located at the site during their years of operation (Misut, 2011). Environmental studies by the U.S. Department of the Interior and the U.S. Geological Survey, at the request of the U.S. Environmental Protection Agency (EPA), confirm the following contaminants in the groundwater: Trichloroethylene, Tetrachlorethylene, 1,4-Dioxane, unregulated contaminants and approximately 20 more chemicals that contain radium, a toxic element that contributes to adverse human health effects. A number of maps composed by the EPA and the U.S. Department of Environmental Conservation are available showing different scenarios of the contamination, including predictions of future groundwater plume spread (US-EPA; 2017). Our research is focused on the following two major questions: How informed are the surrounding communities regarding the actual levels of groundwater contamination and are communities and local entities satisfied by Navy's and Grumman's efforts to clean up contamination? A statistical analysis of water quality and groundwater contamination awareness was conducted utilizing a survey of restaurant businesses in Bethpage, Hicksville and Farmingdale, NY (see Figure 1). These communities were chosen based on their location in proximity to the initial area of the groundwater contamination at Northrop Grumman.

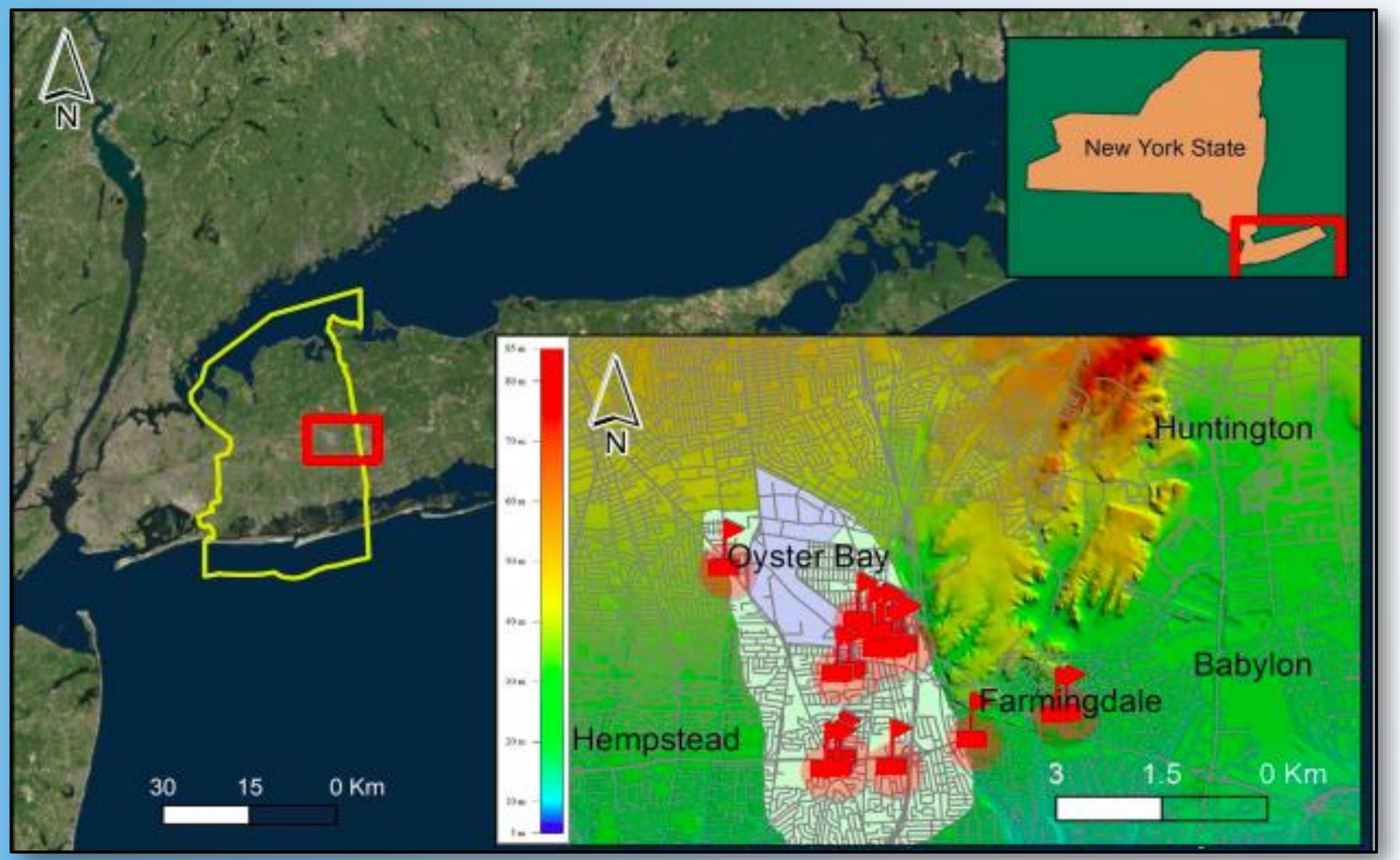

Figure 1: Geographical map of the locations Bethpage, Hicksville, and Farmingdale communities in Nassau County (yellow polygon). The light blue shaded polygon area outlines the Northrop Grumman property, and the light green shading depicts the extent of the plume as of 2012 (after Long Island Express, 2016). Questionnaire survey was conducted, and 21 anonymous responses were collected (shown by red flags) by Hofstra University.

Methods

- A questionnaire survey was distributed to businesses directly connected to food service in the communities of Bethpage, Hicksville and Farmingdale, New York to collect data on their viewpoint of the quality of their water and how aware they are of the Northrop Grumman groundwater plume.
- Over 34 surveys (10-question survey) were distributed through email, phone calls and in-person visits to food service businesses. 21 surveys were completed mostly by in-person visits. A Likert-scale model (Allen & Seaman, 2007) was employed, with responses ranging from 1-5, with 1 corresponding to disagree and 5 corresponding to agree.
- Questions 1-4 and 5-9 were combined into categorical variables to better compare the results; questions 1-4 focusing on the quality of water and questions 5-9 focusing on the awareness of the Northrop Grumman plume. Question 10 was considered its own categorical variable.
- The data was then standardized (percent) to statistically analyze results. When completing the Chi-square test, we re-coded the variables to increase the significance in the results and to better understand water quality and contamination awareness. Agree was coded as 1, neither agree nor disagree as 3 and disagree as 0.

Results

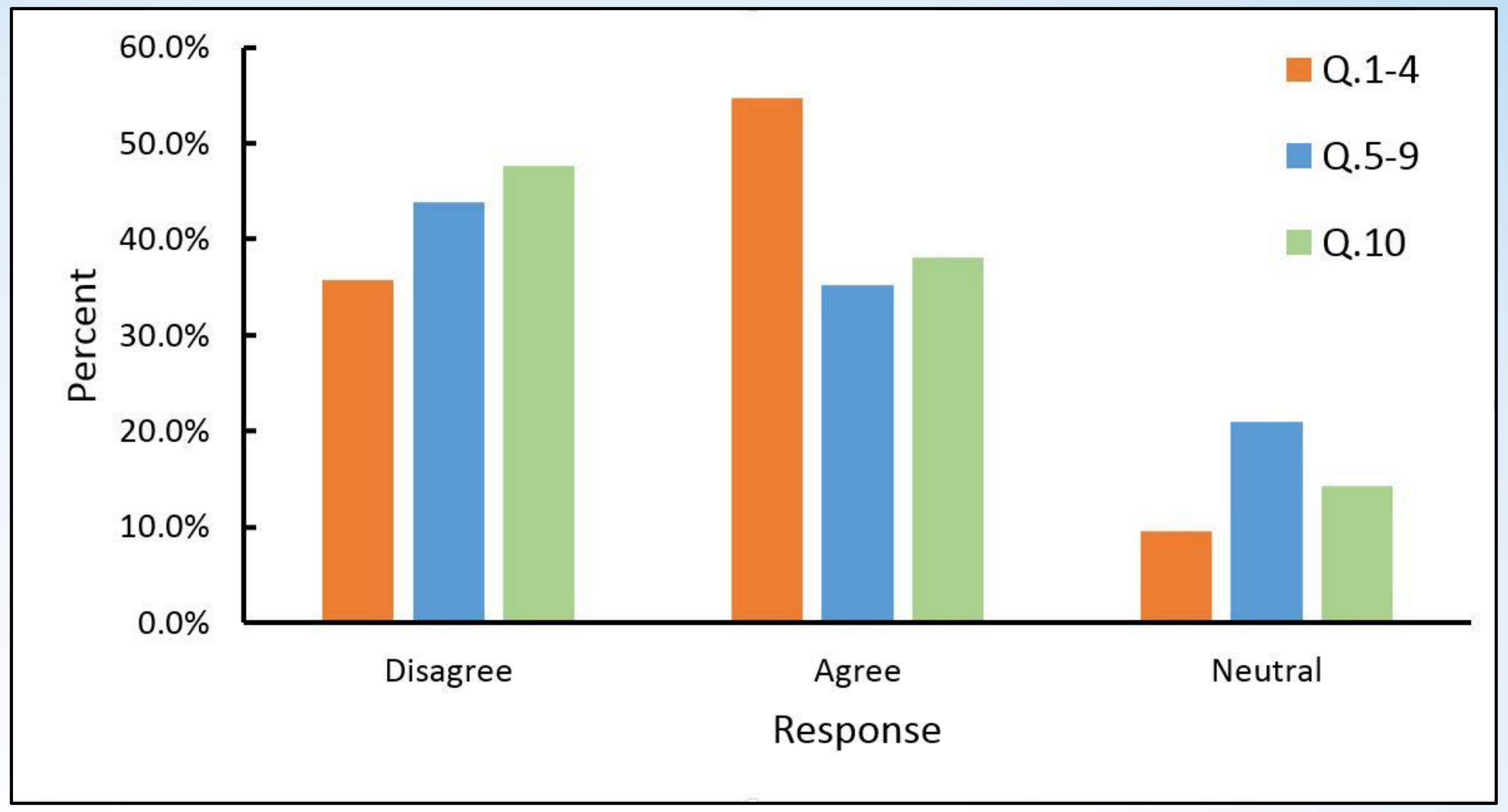

Figure 2: Chart of correlation results of questionnaire survey where questions have been reclassified into three categories. Questions 1-4 depicted by the orange bar represents perceptions of water quality in area. Questions 5-9 depicted by the blue bar represents the level of awareness of the Northrop groundwater plume and question 10, depicted by the green bar, represents responses of how efficient business owners feel about the quality of water and how aware they are about the Grumman plume.

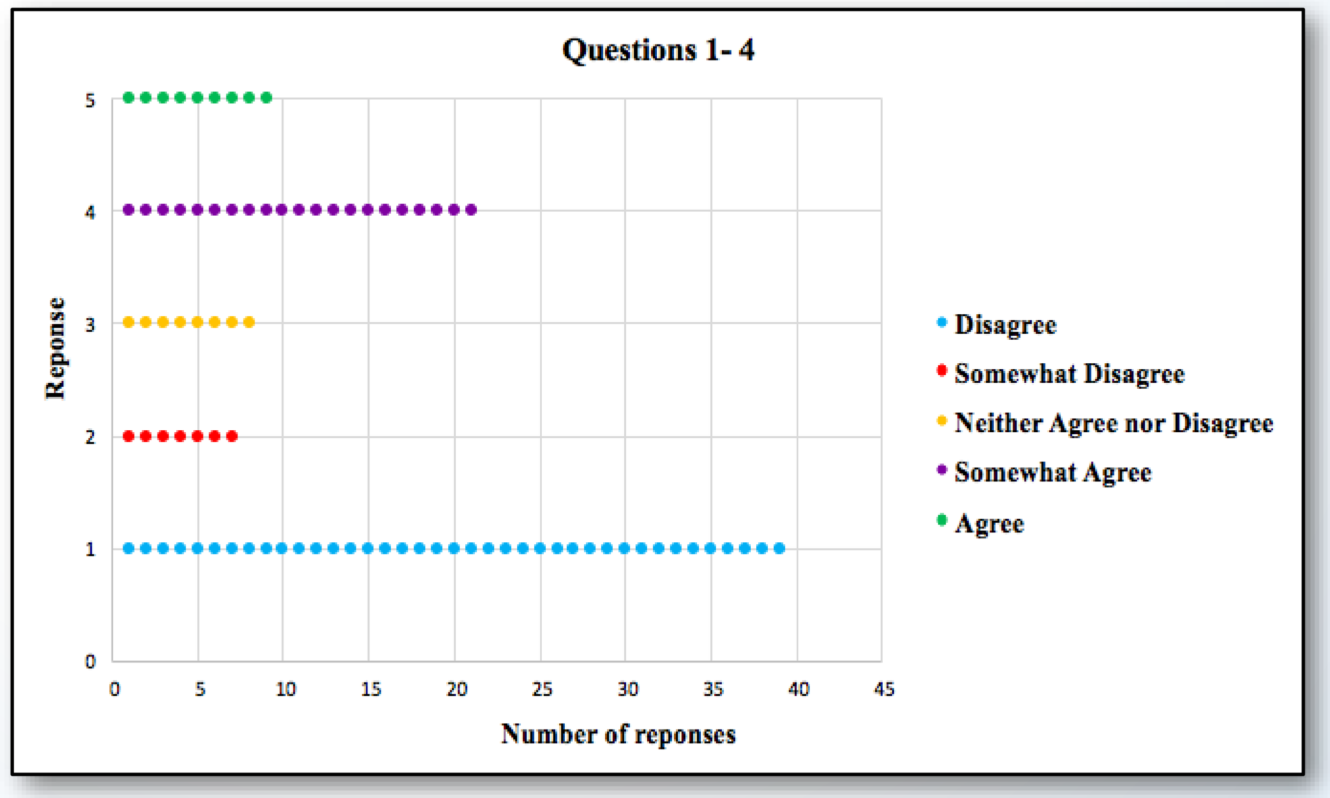

Figure 3: Aggregation of answers from questions 1-4 depict the business perception of water quality within the geographical test area.

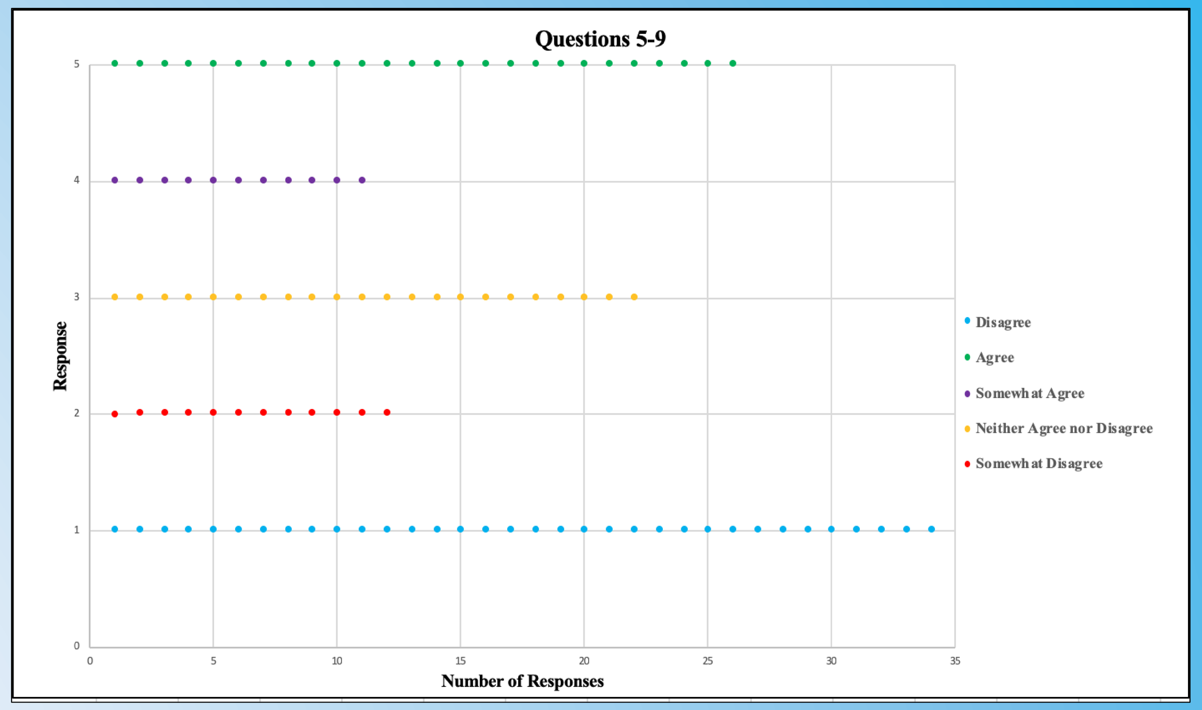

Figure 4: Aggregation of answers from questions 5-9 indicating awareness of Northrop Grumman Plume within the geographical test area.

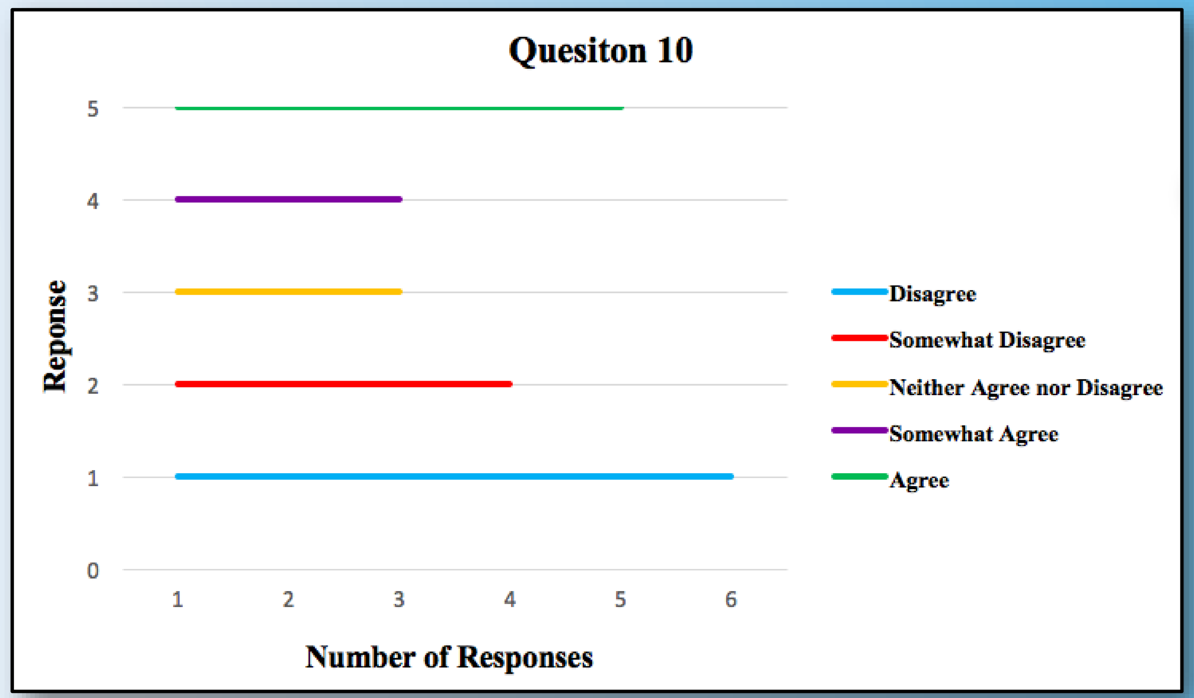

Figure 5: Aggregation of answers from questions 10, corresponding to the level of confidence for communicating environmental emergencies via telephone notification.

Discussion

- 54.8% of the 21 respondents agreed for our combined variable (Q.1-4), corresponding to perception of water quality, while 35.7% of the respondents disagreed.
- The aggregated responses from questions 5-9 show 43.8% of the respondents disagreed, corresponding to level of awareness of the Northrop groundwater plume, while 35.2% of the respondents agreed. For the third variable (Q.10), 47.6% of respondents disagreed, corresponding to the level of confidence in methods of notification for environmental emergency, while 38.1% of respondents agreed.
- Only three of the Chi-Square tests, questions 1 and 2; 1 and 4 ; and 6 and 3, provided a Cramer's V of .58 or higher, resulting in a moderate correlation. The sole outlier Chi-Square test, corresponding to questions 8 and 9, had a high correlation with a Cramer's V of .813. Figure 2 represents the correlation results of the questionnaire survey.

Acknowledgement:

The ten-question survey was submitted to and approved by the Hofstra Institutional Review Board (IRB) for the use of humans as research subjects. Responses to surveys are anonymous and cannot be traced back to the respondent or their business and are completely confidential. Responses were combined with those of many others and summarized in a report to further protect anonymity. We also would like to thank Hofstra University and especially the chair of the department of Geology, Environment and Sustainability for the 2019 conference expenses.

Conclusion

The survey study of the level of awareness of the Northrop Grumman contamination plume in the surrounding communities of Bethpage, Hicksville and Farmingdale, NY indicates that business owners in these communities are aware of the local contamination. However, they do not recognize the severity and impact of the groundwater quality. The analysis of responses related to the level of awareness of the extent of the plume have a neutral response, demonstrating that business owners are not appropriately concerned with gaining greater awareness of the growing contamination. We can conclude that most community businesses are moderately aware of the plume and its impact on groundwater quality, while others are not concerned that this contamination has occurred and affected their business property.

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